

JUNSOO HA

Seoul, Korea

📧 [kuc2477](#) ✦ [in](#) [kuc2477](#) ✦ [✉ junsoo.ha.contact@gmail.com](#)

RESEARCH INTERESTS

Algorithmic Game Theory, Deep Learning Theory, AI Alignment & Safety

EDUCATION

Seoul National University

Master of Science in Artificial Intelligence

Advisor: Prof. Gunhee Kim

Seoul, Korea

Sep. 2020 - Aug. 2023

Hanyang University

Bachelor of Science in Software

Bachelor of Science in Mathematics

Graduated with High Distinction

Seoul, Korea

Mar. 2013 - Aug. 2020

GPA: 4.08 / 4.5

PUBLICATIONS

Stochastic Gradient Descent Ascent is Suboptimal for Nonconvex-PL Min-Max Games

Junsoo Ha

Preprint, 2026

A Horizon-Independent Regret Bound for Optimistic Hedge in General-Sum Games

Junsoo Ha

Preprint, 2026

A Logarithmic Regret Bound for Optimistic Hedge in General-Sum Games

Junsoo Ha

Preprint, 2026

High-Probability Convergence Guarantees of Stochastic Gradient Descent Ascent in Structured Nonconvex Min-Max Games

Junsoo Ha

Conference on Learning Theory (COLT), 2026

On Convergence of Lookahead in Smooth Games

Junsoo Ha, Gunhee Kim

International Conference on Artificial Intelligence and Statistics (AISTATS), 2022

A Neural Dicitale Process Mixture Model for Task-Free Continual Learning

Soochan Lee, Junsoo Ha, Dongsu Zhang, Gunhee Kim

International Conference on Learning Representations (ICLR), 2020

Harmonizing Maximum Likelihood with GANs for Multimodal Conditional Generation

Soochan Lee, Junsoo Ha, Gunhee Kim

International Conference on Learning Representations (ICLR), 2019

RESEARCH & INDUSTRIAL EXPERIENCE

Unknot Inc.

Quantitative Research Scientist

Built a flexible bespoke research framework that supports any existing financial markets

Built a highly parallelized, error-tolerant production inference system

Seoul, Korea

Aug. 2021 - Aug. 2024

Redesigned a data pipeline from scratch to support thousands of assets across the world
Developed a foundation model that could adapt to multiple financial markets
Raised code quality standards and introduced automated tests to make research reliable

SNU Vision & Learning Lab.

Research Assistant

Seoul, Korea

Nov. 2017 - Aug. 2021

Buzzni

Backend Software Engineer

Seoul, Korea

Mar. 2017 - Jul. 2017

Geopia

Full-Stack Software Engineer

Seoul, Korea

Apr. 2015 - Feb. 2017

Nexol System

Software Engineering Intern

Seoul, Korea

Jan. 2015 - Mar. 2015

LG Electronics

Software Engineering Intern

Seoul, Korea

Jul. 2014 - Aug. 2014

HONORS & AWARDS

Samsung Human-Tech Paper Award

2019

Samsung Electronics

2nd place; Silver Prize

Distinguished Service Award

2019

Hanyang University

RELEVANT COURSEWORKS

Graduate Convex Optimization Theory, Measure Theory

Undergraduate Deep Learning, Mathematical Statistics I-II, Abstract Algebra I-II, Topology I-II, Linear Algebra II, Ordinary Differential Equations, Advanced Calculus I, Vector Calculus, Differential Geometry, Elementary Hyperbolic Geometry

SKILLS

Machine Learning

PyTorch, Tensorflow

Python

Django, Flask, SQLAlchemy, Celery

JavaScript

ES6/7, React, Redux, Electron, Immutable

Database

MySQL, PostgreSQL, SQLite, Redis